QUASI-COHERENT MODULES ON A NOETHERIAN AFFINE FORMAL ALGEBRAIC SPACE NEED NOT FORM AN ABELIAN CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author's footnote on the definition of quasi-coherent is part of the counted statement and is kept verbatim. The final 'this category is not abelian' of the context points at Section non-abelian, whose kernel/cokernel machinery is included as the first context span; the witnessing non-exactness is the delivered card L48306.

Editorial note. The label lemma-quasi-coherent-not-abelian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 792-805 (section-non-abelian): for R a ring and I finitely generated, the category A of I-adically complete R-modules has cokernel of phi: M -> N given by the I-adic completion of the usual cokernel (Algebra, hathat-finitely-generated) and kernel the usual kernel, which is closed and I-adically complete by Lemma closed-is-complete (delivered card L48284)
- Context span 1161-1201 (section-nonabelian-QCoh): X = Spf(Z_p) with the p-adic topology is a sheaf in sets on (Sch/Z)_fppf and gives a stack in setoids (Stacks, when-stack-in-sets); quasi-coherent modules on it correspond to systems (M_n) of abelian groups annihilated by p^n with M_n = M_{n+1}/p^n M_{n+1}, equivalently to the p-adically complete abelian group M = lim M_n with M_n = M/p^n M (Algebra, limit-complete); the category of quasi-coherent modules on X is thus equivalent to the category of p-adically complete abelian groups, which the context states is not abelian, see Section non-abelian

Source reference. Original definition 0317: sources/stacks-chow/tags.txt, source lines 3974–3974. Prerequisite pointer invoked at 1222; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Prerequisite pointer invoked at 1227; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09B8: sources/stacks-chow/tags.txt, source lines 11786–11786. Prerequisite pointer invoked at 1268; it names a standard fact of a

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sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0AIF: sources/stacks-chow/tags.txt, source lines 13263–13263. Prerequisite pointer invoked at 1244; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0HB3: sources/stacks-examples/pinned-original-source.tex, source lines 808–808. Prerequisite pointer invoked at 1229; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07JQ: sources/stacks-examples/pinned-original-source.tex, source lines 790–790. Prerequisite pointer invoked at 1272; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0430: sources/stacks-chow/tags.txt, source lines 5347–5347. Prerequisite pointer invoked at 1247; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06WK: sources/stacks-chow/tags.txt, source lines 8819–8819. Prerequisite pointer invoked at 1255; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 06WF: sources/stacks-chow/tags.txt, source lines 8814–8814. Prerequisite pointer invoked at 1232; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 06WU: sources/stacks-chow/tags.txt, source lines 8828–8828. Prerequisite pointer invoked at 1236; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let R be a ring and let $I \subset R$ be a finitely generated ideal. Consider the category $\mathcal{A}$ of I-adically complete R-modules, see Algebra, Definition 0317. Let $\varphi : M \to N$ be a morphism of $\mathcal{A}$. The cokernel of φ in $\mathcal{A}$ is the I-adic completion $(\mathrm{Coker}(\varphi))^\wedge$ of the usual cokernel (as I is finitely generated this completion is complete, see Algebra, Lemma 05GG). Let $K = \mathrm{Ker}(\varphi)$; since φ is continuous this is a closed submodule of M. By Lemma 0HB3 below we see that K is I-adically complete and hence is the kernel of φ in $\mathcal{A}$.

In Sheaves on Stacks, Section 06WF we defined the category of quasi-coherent modules on a category fibred in groupoids over Sch. Although we show in Sheaves on Stacks, Section 06WU that this category is abelian for algebraic stacks, in this section we show that this is not the case for formal algebraic spaces.

Namely, consider $\mathbf{Z}_p$ viewed as topological ring using the p-adic topology. Let $X = \mathrm{Spf}(\mathbf{Z}_p)$, see Formal Spaces, Definition 0AIF. Then X is a sheaf in sets on $(Sch/\mathbf{Z})_{fppf}$ and gives rise to a stack in setoids $\mathcal{X}$, see Stacks, Lemma 0430. Thus the discussion of Sheaves on Stacks, Section 06WU applies.

Let $\mathcal{F}$ be a quasi-coherent module on $\mathcal{X}$. Since $X = \operatorname{colim} \mathrm{Spec}(\mathbf{Z}/p^n\mathbf{Z})$ it is clear from Sheaves on Stacks, Lemma 06WK that $\mathcal{F}$ is given by a sequence $(\mathcal{F}_n)$ where

(1) $\mathcal{F}_n$ is a quasi-coherent module on $\mathrm{Spec}(\mathbf{Z}/p^n\mathbf{Z})$, and
(2) the transition maps give isomorphisms $\mathcal{F}_n = \mathcal{F}_{n+1}/p^n\mathcal{F}_{n+1}$.

Converting into modules we see that $\mathcal{F}$ corresponds to a system (M_n) where each M_n is an abelian group annihilated by p^n and the transition maps induce isomorphisms $M_n = M_{n+1}/p^n M_{n+1}$. In this situation the module $M = \lim M_n$ is a p-adically complete module and $M_n = M/p^n M$, see Algebra, Lemma 09B8. We conclude that the category of quasi-coherent modules on X is equivalent to the category of p-adically complete abelian groups. This category is not abelian, see Section 07JQ.

Lemma 0.1. *The category of quasi-coherent*[1] *modules on a formal algebraic space X is not abelian in general, even if X is a Noetherian affine formal algebraic space.*

Proof. See discussion above. □

[1]With quasi-coherent modules as defined above. Due to how things are setup in the Stacks project, this is really the correct definition; as seen above our definition agrees with what one would naively have defined to be quasi-coherent modules on $\mathrm{Spf}(A)$, namely complete A-modules.